

MAT Software Solutions : Version Control Software

Backend Documentation : Valid as of 19 March 2026

Contents:

main.py - p. 2

ConsoleFunctions.py - p. 3 -> p.4

DBFunctions.py -p.6 ->p. 7

main.py :

main.py is the launch file for the MAT Software Solutions Version Control Software

Step 1. On program launch, sets every user in the DB's "loginStatus" field to false.

*The main variable that gets passed around in this program is the "localUser" variable.

Step 2. The program will initialize the "localUser" variable through the "consoleUserLogin" function, passing a string of value "dummyInput". (see page !!!!!!!! for further documentation). If the "localUser" variable is initialized as an integer value equal to 0, the program will exit with an exit code of 500

Step 3. The program will enter its main loop, where all further functionality lives

```
while True:
    if localUser != "dummyInput" and localUser != 0 :
        uiParagraph("You are currently logged in as : " + localUser.username)
        localUser = mainFuncs[consoleMenu("basicUserMenu")](localUser)
    else : localUser = consoleUserLogin("dummyInput")
```

At the start of every iteration, the loop will check if the "localUser" variable has not been set to equal a string of value "dummyInput" or an integer of value 0

If the program has detected neither is true, it will call the uiParagraph function to display which user is currently logged in. (see page !!!!!!!! for uiParagraph documentation) . The program will then execute the following line of code :

```
localUser = mainFuncs[consoleMenu("basicUserMenu")](localUser)
```

TL;DR :

A menu is displayed in the console, prompting the user to decide if they want to either Log Out, or open the File Manager. In either case, the localUser variable is passed to either of the following functions :

consoleUserLogout ; consoleFileManager

Definitions :

mainFuncs - a map that takes a string as input and returns a function ;

consoleMenu : see p.2 for consoleMenu documentation

ConsoleFunctions.py :

ConsoleFunctions.py is a file containing functions that access Version Control functionality through a console window or terminal.

TL;DR :

consoleUserLogin - takes an input called "inputUser" which is only expected to be either a string of value "dummyInput" or an actual User variable.

The program will then prompt the user to log in with their credentials. If an incorrect login is detected, consoleUserLogin will return 0. If logging in was successful and inputUser is detected as NOT being a string of value "dummyInput", the program will call the function "userLogout", passing the inputUser variable (see p. 6 for userLogout documentation) ; The idea here is if a User A is logging in as User B, to log out User A before continuing program code;

consoleUserLogout - takes an input called "inputUser" which is only expected to be a User variable. If inputUser leads to a User, whose loginStatus in the DB is 1 (logged in), then consoleUserLogout will set that to 0 (logged out). If inputUser leads to a User, whose loginStatus in the DB is 0 (logged out), consoleUserLogout will do nothing. In either case, consoleUserLogout returns a string of value "dummyInput"

consoleUserProjectList - ###QUESTIONS TO DB DEVS

consoleUserAuditLog - takes an input called "inputUser" which is only expected to be a User variable. Uses a loop to print all Audit Logs from DB that are associated with inputUser, sorted by project, using the uiLogs function (see p. 5 for uiLogs documentation), as well as the getUserProjects and getProjectAuditLogs functions (see p. 6 for getUserProjects as well as getProjectAuditLogs documentation)

basicUserMenu - an array containing strings, which are later printed as menu options

menuMap - a map that takes a string as input and returns the associated array of options

consoleMenu - a function that takes an input called "inputMenuType", which is expected to be a string. The idea around consoleMenu is to have a unified way to display different menu options and return user input, and that is all it does. Currently, the only menu accessible through the consoleMenu function is "basicUserMenu". In all cases, the consoleMenu function returns a string (currently, the only two options being "Log Out", and "File Manager"). The only call to the consoleMenu function at the moment is in main.py (see p.1 for main.py documentation)

ConsoleFunctions.py:

consoleFileManager - a function that takes an input called "inputUser" that is only expected to be an actual User variable.

This is the current location of all DB functions that are accessible through a console window or terminal. The user is prompted to pick an option out of a current 11, and his input is further redirected accordingly. The current implementation uses a simple if/elif system to filter user input, which will certainly change in the future. It is possible that in the future the consoleFileManager function uses the consoleMenu function (documented above) to later filter user input through a map, as is done in main.py (see p.1 for main.py documentation) with the "mainFuncs" map (see p.1 for mainFuncs documentation)

see p.6 ->p.7 for all DBFunctions documentation

ConsoleUI.py :

ConsoleUI.py is a file containing functions that help reduce print statements in other parts of the program

clearConsole - a function that prints 100 lines of /n

printCharLine - a function that prints 100 characters of its input

uiError - a function that prints its input as an error type that MAT VCS ran into

uiParagraph - a function that prints its input between two lines of "=" by calling printCharLine (documented above)

uiFiles - a function that takes an array of type VersionFile and prints every element's .path and .content fields

uiLogs - a function that takes an array of type AuditLog and prints every element's .timestamp, .user.username, .action, and .details fields ###QUESTIONS FOR DB DEVS

uiProjectVersion - a function that takes an input of type ProjectVersion and prints its .created_at, .version_number, and .status fields

uiFileManagerMenu - a function that prints all the available options in the consoleFileManager function (see p.4 for consoleFileManager documentation)

uiUserInputPrompt - a function that takes an input of type string, prints that string, awaits user input and upon receiving it calls clearConsole (documented above) and returns userInput

uiUserInputPromptNoCLS - a function identical to the uiUserInputPrompt function (documented above), with the difference of not calling clearConsole (documented above)

DBFunctions.py :

DBFunctions.py is a file that contains Django-level function calls that directly alter the DB

getUserAuditLogs - a function that takes an input called inputUserID, which is expected to be an integer and returns a queryset of all AuditLog DB entries associated with that inputUserID

getProjectAuditLogs - a function that takes an input called inputProjectID, which is expected to be an integer and returns a queryset of all AuditLog DB entries associated with that inputProjectID

getProjectByID - a function that takes an input called inputProjectID, which is expected to be an integer and returns a variable of type Project, which is a project, whose PK is equal to inputProjectID

getProjectByVersionFile - a function that takes an input called inputVersionFile, which is expected to be of type VersionFile and returns a variable of type Project, which is the project which corresponds to inputVersionFile's associated ProjectVersion
##QUESTIONS FOR DB DEVS

getUserProjects - a function that takes an input called "user" which is expected to be an actual User variable, and returns a queryset of all projects associated with the user variable
###QUESTIONS FOR DB DEVS

getProjectVersionsByProjectID - a function that takes an input called "inputProjectID" which is expected to be an integer, and returns a queryset of all ProjectVersion entries in the DB associated with inputProjectID

getProjectVersionSpecificByVersionFile - a function that takes an input called "inputVersionFile" which is expected to be an actual VersionFile variable, and returns a single ProjectVersion

getAllProjectFiles - a function that takes an input called "inputProjectId" which is expected to be an integer, and returns a queryset of all VersionFile entries in the DB associated with inputProjectId

getOrCreateUser - a function that gets or creates a user based on input data

logCreateProject - a function that creates an AuditLog of project creation based on input data

addProjectToDB - a function that adds a Project entry to the DB based on input data and calls logCreateProject (documented above)

DBFunctions.py :

logCreateProjectVersion - a function that creates an Audit Log of ProjectVersion creation based on input data

addNextProjectVersionToDB - a function that adds a ProjectVersion entry to the DB based on input data and calls logCreateProjectVersion (documented above) ###QUESTIONS FOR DB DEVS

logCreateVersionFile - a function that creates an Audit Log of VersionFile creation based on input data

addVersionFileToDB - a function that adds a VersionFile entry to the DB based on input data and calls logCreateVersionFile (documented above)

logRemoveVersionFile - a function that creates an Audit Log of VersionFile removal based on input data

removeVersionFileFromDB - a function that removes an entry of VersionFile from the DB based on input data and calls logRemoveVersionFile (documented above)

removeProjectFromDB - currently unimplemented

removeProjectVersionFromDB - currently unimplemented

userLogOut - a function that takes an input called "inputUser" which is expected to be an actual User variable and sets its DB entry's loginStatus value to 0 (logged out)

userLogIn - a function that takes an input called "inputUser" which is expected to be an actual User variable and sets its DB entry's loginStatus value to 1 (logged in)

approveProjectVersion - a function that sets a ProjectVersion DB entry's status to "APPROVED" based on input data and creates an AuditLog entry of the action

editVersionFileToVersion - currently unimplemented